UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science
Faculty of Computing

GROUP PROJECT

(Benchmarking of Computer Performance)

Programme : Bachelor of Computer Science

Subject Code : SECR1033

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Section : 02

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Video Link :

Date :

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1.0 Introduction

Problem Background

The Central Processing Unit (CPU) is a key component of a computer that serves as its "control center". It interprets and executes instructions from the computer's memory, working with hardware components to complete tasks. Furthermore, the CPU is useful in determining the overall performance and efficiency of a system. Nowadays, new CPU models are always being released with improved architecture, therefore we must understand how to analyze and compare CPU performance in order to determine which brand of computer is the best purchase. CPU benchmarking is one of the tools that may be used in this evaluation process, allowing consumers to evaluate and understand the capabilities and limitations of various CPUs, as well as developers to design better CPUs.

Aim

The primary goal of this research is to provide a thorough introduction to CPU benchmarking, which provides insights into computer performance, hardware variances, and the complexities of software optimization. We will look at different benchmarks and tools for evaluating CPU performance. Furthermore, We'll also run a simple Assembly programme on three different computers and use functions to calculate the difference in CPU running time between the three.

Objective

1. **Understand the Principles of CPU Benchmarking** : Gain a foundational understanding of the concepts, importance and methodologies of CPU benchmarking.

2. **Compare the Performance of Different CPUs** : Evaluate and compare the performance of various CPUs to identify strengths and weaknesses.

3. **Analyse the Impact of Different Workloads on CPU Performance** : Assess how different types of workloads affect CPU performance.

4. **Learn to Use Free Benchmarking Tools and Interpret Their Results** : Acquire practical skills in using free benchmarking tools and learn how to interpret their outputs.

Scopes

Computer Specification

For computer to be benchmarked, we use Acer Aspire (A515-56), HP Laptop 15s-fq5xxx

Specifications	Computer Type		
	Acer Aspire 5	HP Laptop 15s-fq5xxx	ASUS Zenbook S13 OLED
Precessor Type	Intel Core i5 1135G7	Intel Core i7 1255U	AMD Ryzen 7 6800U
RAM Size	8 GBytes	32 GBytes	16 GBytes
GPU Type	Intel® Iris Xe Graphics	Intel® Iris Xe Graphics	AMD Radeon™ Graphics
Motherboard	TGL Iris TL (U3E1)	HP 8A20	AsusTEK COMPUTER INC. UM5302TA
Bus Specs.	PCI-Express 3.0 (8.0 GT/s)	PCI-Express 3.0 (8.0 GT/s)	PCI-Express 4.0 (16.0 GT/s)

Benchmarking Tools Features

1. CPU-Z
   It is a lightweight tool that provides detailed information about the CPU, including its name, architecture, clock speeds, cache, and more. It also offers real-time monitoring of core frequencies and memory performance. The main uses of CPU-Z includes :
   - Detailed System Specifications
   - Real-Time Monitoring
   - Diagnostics

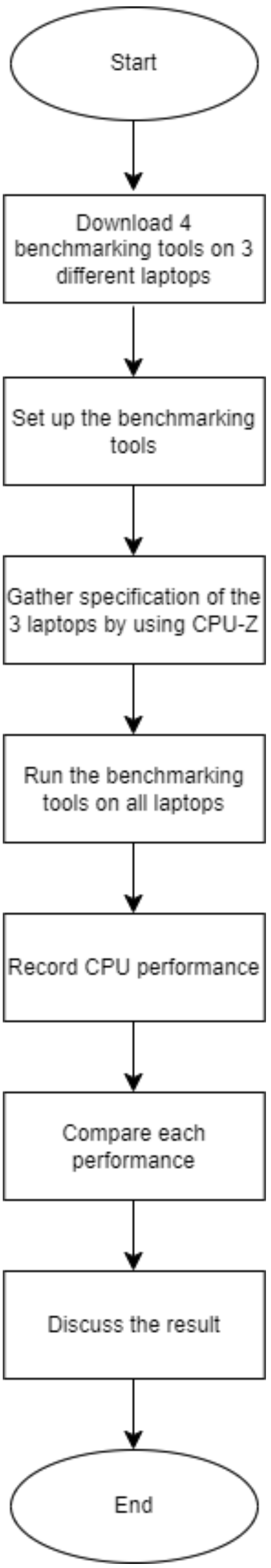
2. Geekbench
   It is a popular benchmarking tool that measures the single-core and multi-core performance of a CPU using a variety of simulated workloads. It provides a score that can be compared across different systems. The main uses of Geekbench includes :
   - Standardized Performance Testing
   - Comparative Analysis
   - Performance Evaluation

3. Task Manager
   The Task Manager in Windows provides real-time monitoring of CPU usage, clock speed, and thermal performance. It is a built-in utility that offers a quick overview of system resource utilization. The main uses of Task Manager includes :
   - Performance Monitoring
   - Process Management
   - Troubleshooting

4. MiniTool Partition Wizard
   While primarily a disk management tool, MiniTool Partition Wizard includes benchmarking features to test disk performance. It provides insights into read and write speeds, which can indirectly impact CPU performance. The main uses of MiniTool Partition Wizard includes :
   - Storage Performance Testing
   - Partition Management
   - System Optimization

2.0 Part 1

2.1 Flowchart of Project Part 1



2.2 Benchmarking Result

CPU-Z

Benchmark	Computer Type		
	Acer Aspire 5	HP Laptop 15s-fq5xxx	ASUS Zenbook S13 OLED
CPU Technology	10nm	10nm	6nm
Number of Core	4	2P+8E	8
Core Speed	1305.12 MHz	4090.00 MHz	1393.61 MHz
Bus Speed	100.39 MHz	99.76 MHz	99.49 MHz
RAM Size	8 GBytes	32 GBytes	16 GBytes
GPU Size	-	-	-

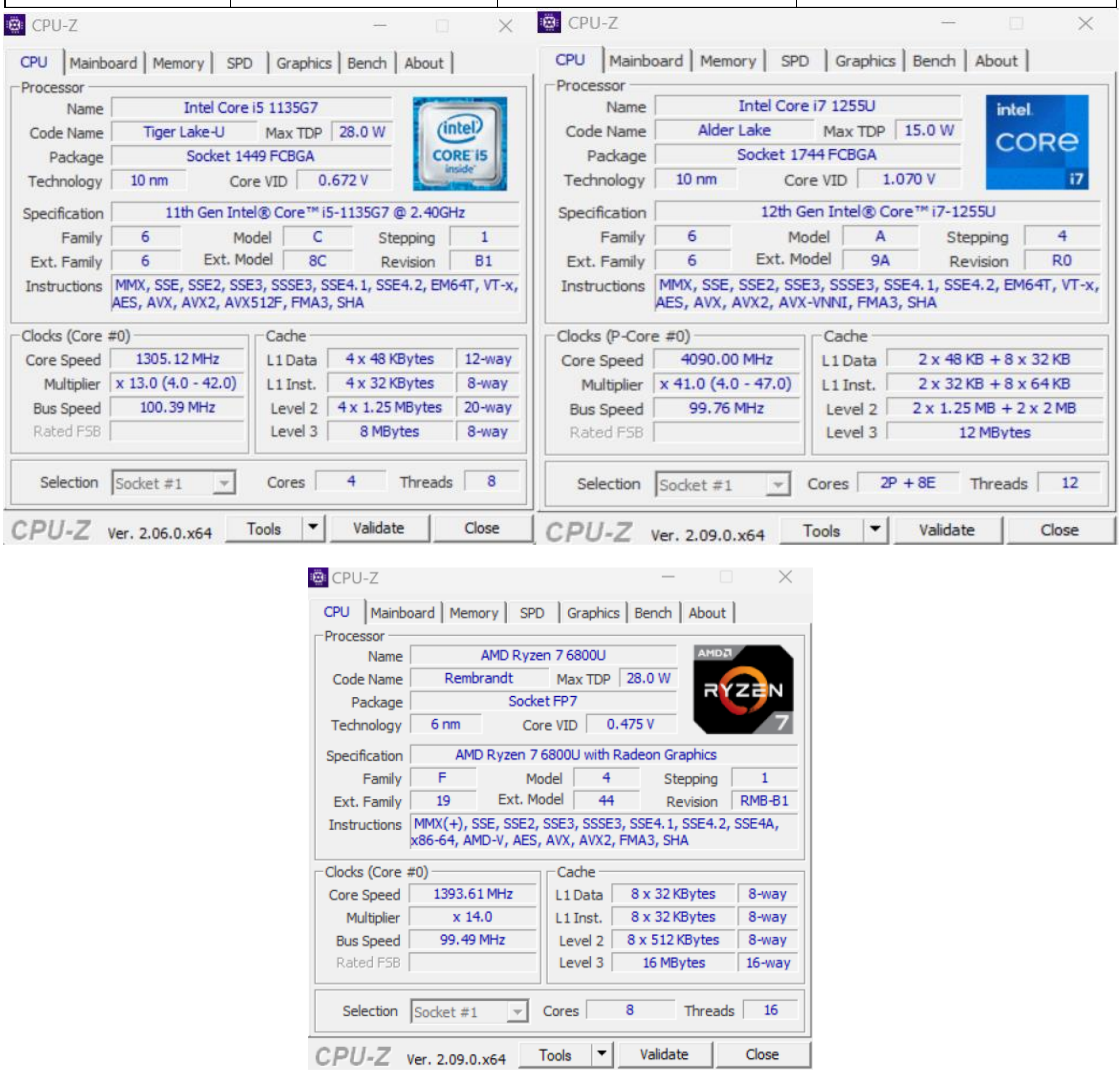
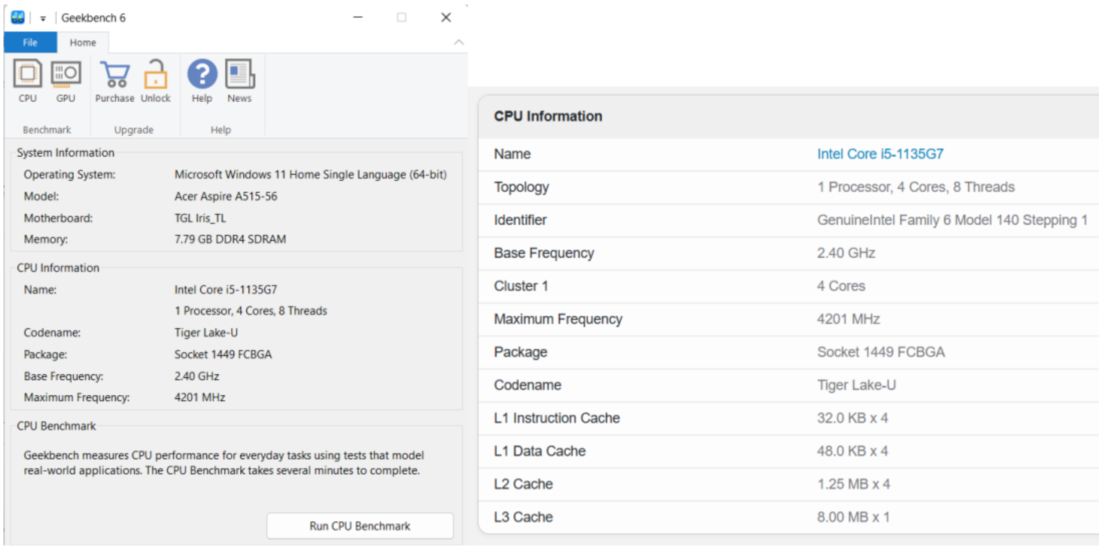


Figure A1: CPU-Z Output from 3 different computer

Geekbench

<table>
<tr><td>Acer Aspire A515-56</td><td>
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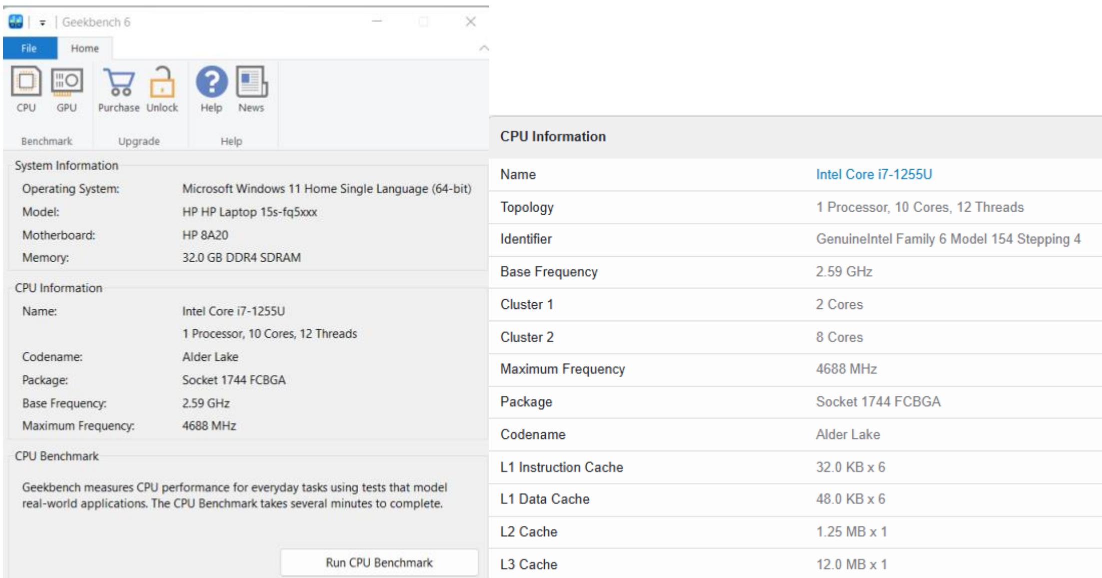
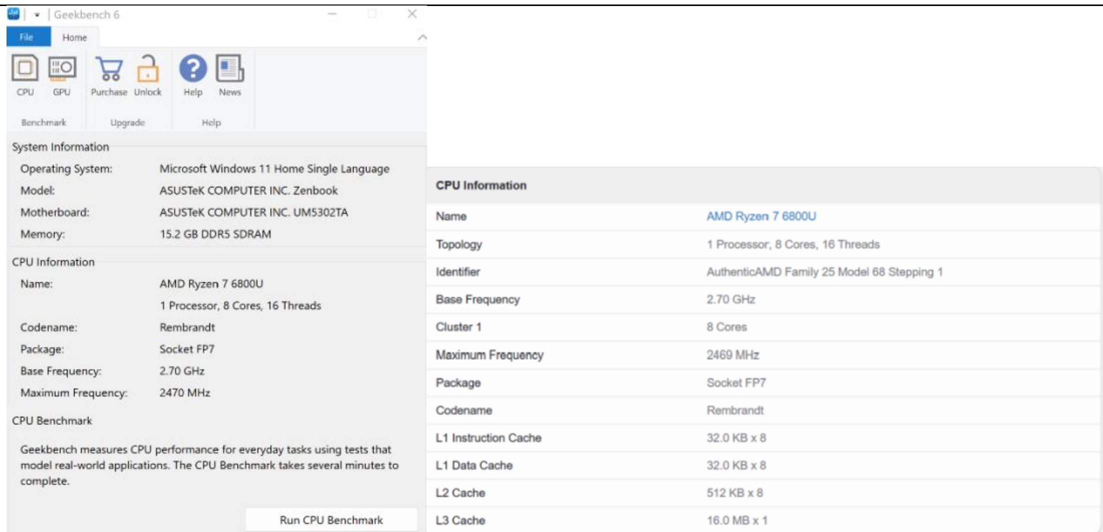
<tr><td>HP Laptop

15s-fq5xxx</td><td>
</td></tr>
<tr><td>ASUS Zenbook S13 OLED</td><td>
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</table>

Table 2: Greekbench Output of 3 different computer

Greekbench -CPU Load Stress Test (CPU Test)

Acer Aspire A515-56

Single-Core Performance

Single-Core Score	1163	
File Compression	1020	146.5 MB/sec
Navigation	1284	7.74 routes/sec
HTML5 Browser	1060	21.7 pages/sec
PDF Renderer	990	22.8 Mpixels/sec
Photo Library	1169	15.9 images/sec
Clang	1041	5.13 Klines/sec
Text Processing	1118	89.5 pages/sec
Asset Compression	1321	40.9 MB/sec
Object Detection	1329	39.8 images/sec
Background Blur	1752	7.25 images/sec
Horizon Detection	1413	44.0 Mpixels/sec
Object Remover	838	64.4 Mpixels/sec
HDR	1313	38.5 Mpixels/sec
Photo Filter	1182	11.7 images/sec
Ray Tracer	888	858.9 Kpixels/sec
Structure from Motion	1301	41.2 Kpixels/sec

Multi-Core Performance

Multi-Core Score	3346	
File Compression	1435	206.1 MB/sec
Navigation	4714	28.4 routes/sec
HTML5 Browser	1855	38.0 pages/sec
PDF Renderer	3524	81.3 Mpixels/sec
Photo Library	4782	64.9 images/sec
Clang	5753	28.3 Klines/sec
Text Processing	1491	119.4 pages/sec
Asset Compression	6585	204.0 MB/sec
Object Detection	1840	55.0 images/sec
Background Blur	5037	20.8 images/sec
Horizon Detection	4759	148.1 Mpixels/sec
Object Remover	3315	254.9 Mpixels/sec
HDR	3712	108.9 Mpixels/sec
Photo Filter	2622	26.0 images/sec
Ray Tracer	5317	5.14 Mpixels/sec
Structure from Motion	4029	127.6 Kpixels/sec

HP Laptop 15s-fq5xxx

Test	Single-Core Performance	Single-Core rate	Multi-Core Performance	Multi-Core rate
Single-Core Score / Multi-Core Score	2029		6005	
File Compression	1604	230.3 MB/sec	3851	553.1 MB/sec
Navigation	1620	9.76 routes/sec	7982	48.1 routes/sec
HTML5 Browser	1893	38.8 pages/sec	7365	150.8 pages/sec
PDF Renderer	1826	42.1 Mpixels/sec	7623	175.8 Mpixels/sec
Photo Library	1800	24.4 images/sec	6138	83.3 images/sec
Clang	2144	10.6 Klines/sec	6046	29.8 Klines/sec
Text Processing	2141	171.4 pages/sec	2152	172.3 pages/sec
Asset Compression	2072	64.2 MB/sec	8111	251.3 MB/sec
Object Detection	2184	65.4 images/sec	3785	113.3 images/sec
Background Blur	2902	12.0 images/sec	6796	28.1 images/sec
Horizon Detection	2794	86.9 Mpixels/sec	8236	256.3 Mpixels/sec
Object Remover	1926	148.1 Mpixels/sec	5873	451.6 Mpixels/sec
HDR	2278	66.9 Mpixels/sec	6183	181.4 Mpixels/sec
Photo Filter	2504	24.8 images/sec	6385	63.4 images/sec
Ray Tracer	1877	1.82 Mpixels/sec	9599	9.29 Mpixels/sec
Structure from Motion	1890	59.9 Kpixels/sec	8002	253.4 Kpixels/sec

ASUS Zenbook S13 OLED

Test	Single-Core Performance	Single-Core rate	Multi-Core Performance	Multi-Core rate
Single-Core Score / Multi-Core Score	1102		6125	
File Compression	1195	171.8 MB/sec	4478	643.0 MB/sec
Navigation	1059	6.38 routes/sec	7931	47.8 routes/sec
HTML5 Browser	1111	22.7 pages/sec	6788	139.0 pages/sec
PDF Renderer	1079	24.9 Mpixels/sec	7722	178.1 Mpixels/sec
Photo Library	1000	13.6 images/sec	6082	82.5 images/sec
Clang	1065	5.25 Klines/sec	9091	44.8 Klines/sec
Text Processing	1055	84.5 pages/sec	1363	109.2 pages/sec
Asset Compression	1119	34.7 MB/sec	10096	312.8 MB/sec
Object Detection	586	17.5 images/sec	2792	83.5 images/sec
Background Blur	1545	6.39 images/sec	7724	32.0 images/sec
Horizon Detection	1484	46.2 Mpixels/sec	8975	279.3 Mpixels/sec
Object Remover	1110	85.4 Mpixels/sec	5548	428.5 Mpixels/sec
HDR	1296	38.0 Mpixels/sec	7776	228.2 Mpixels/sec
Photo Filter	1478	14.7 images/sec	5917	58.7 images/sec
Ray Tracer	1002	969.4 Kpixels/sec	11648	11.3 Mpixels/sec
Structure from Motion	1220	38.6 Kpixels/sec	7910	250.4 Kpixels/sec

Table 3: Greekbench CPU Load Stress Test (CPU Test) of 3 different computer

Window Task Manager

<table>
<tr><th rowspan="2">Computer Type</th><th colspan="3">Workloads</th></tr>
<tr><th>High CPU Load</th><th>Mixed Load</th><th>Low CPU Load</th></tr>
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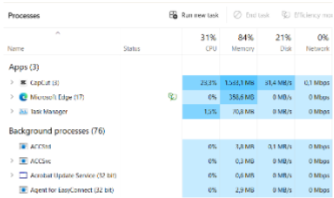
<td>
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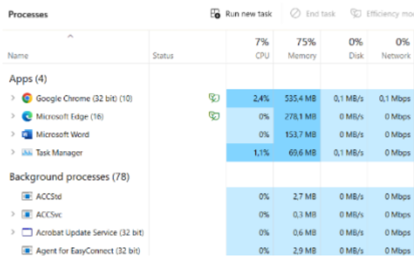
<td>
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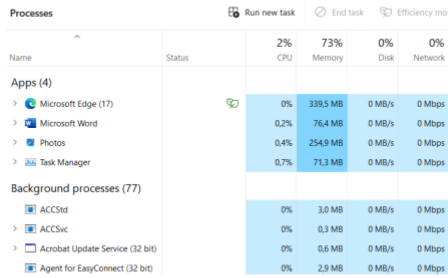
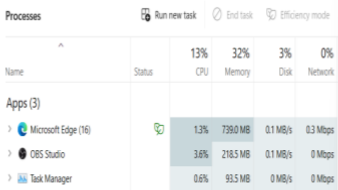
<td>
</td></tr>
<tr><td>HP Laptop
15s-
fq5xxx</td>
<td>
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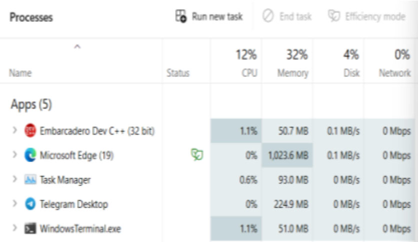
<td>
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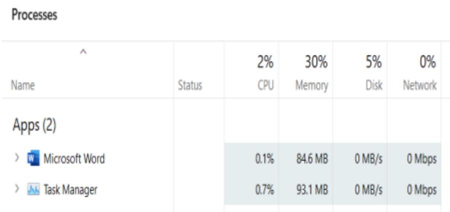
<td>
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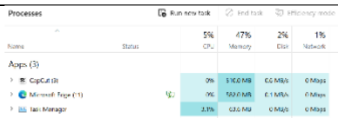
<tr><td>ASUS Zenbook S13 OLED</td>
<td>
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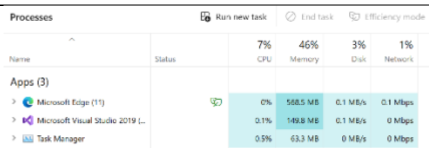
<td>
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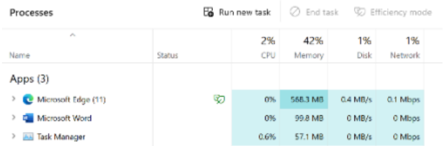
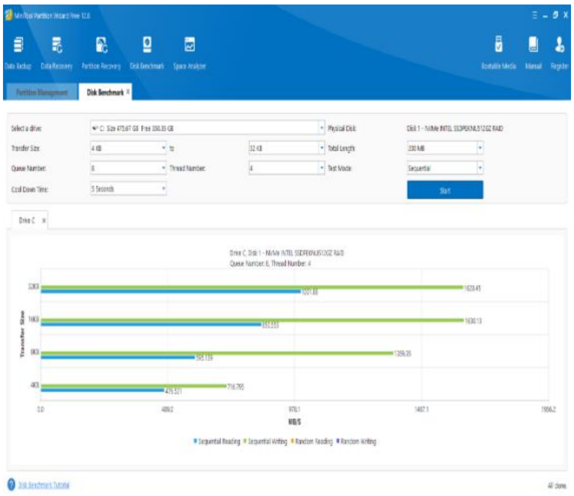
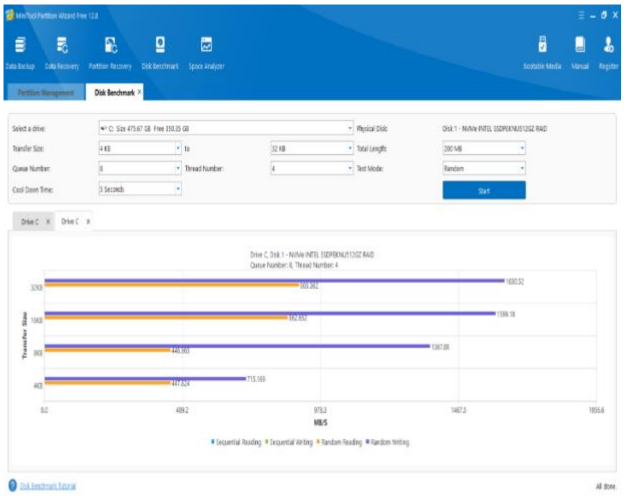
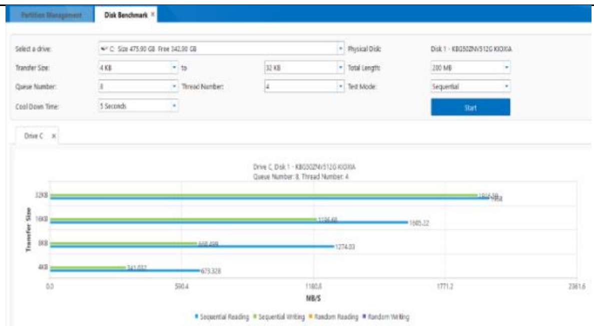
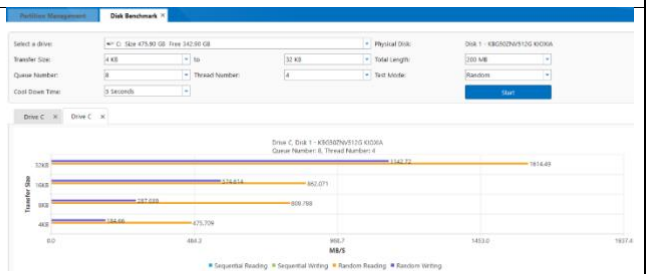
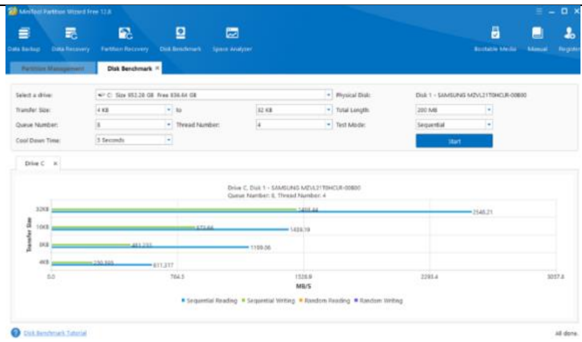
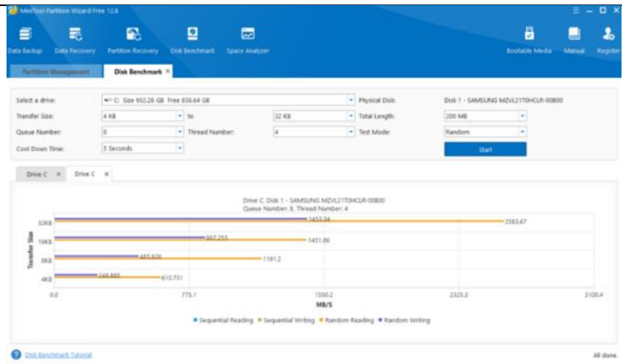
<td>
</td></tr>
</table>

Summarize	High CPU workloads, like video rendering, use a lot of processing power, making the CPU work hard and get hot. Memory and disk use are high, and network use can be heavy if transferring large data.	Mixed workloads, like compiling code while running other applications, have varying CPU demands, causing fluctuating CPU use. Memory and disk use are moderate with some spikes. Network use varies, balanced but can be heavy at times.	Low CPU workloads, like basic office tasks, use little processing power, keeping the CPU cool and responsive. Memory and disk use are low, and network use is minimal, making the system quick and efficient.

Table 4: Process versus resources usage (CPU,Memory,Disk,Network,GPU) of 3 different computer

MiniTool Partition Wizard (Windows)

Computer Types	Mode Test	
	Sequential	Random
Acer Aspire A515-56		
HP Laptop 15s-fq5xxx		
ASUS Zenbook S13 OLED		

2.3 Benchmark Comparison

Computer	CPU-Z (System Info)	Geekbench (CPU stress test)	Task Manager (Active Program and Resource Adjustment)	MiniTools (Disk benchmark)
Acer Aspire A515-56	**CPU Technology :** 10nm **Number of Cores**: 4 **Core Speed :** 1305.12 MHz **Bus Speed :** 100.39 MHz **RAM Size :** 8 GBytes	**Single - Core :** 1163 **Multi - Core :** 3346	**High CPU Load:** CPU: 31% Memory: 84% Disk: 21% Network: 0% **Mixed Load:** CPU: 7% Memory: 75% Disk: 0% Network: 0% **Low CPU Load:** CPU: 2% Memory: 73% Disk: 0% Network: 0%	**Sequential Read** Performance: Acer Aspire A515-56 shows the lowest sequential read speeds across all block sizes, indicating it might be less efficient for read-intensive tasks. **Sequential Write** Performance: Acer Aspire A515-56 demonstrates strong sequential write performance at higher block sizes but falls behind HP for smaller sizes. **Random Read** Performance: Acer Aspire A515-56 lags in random read speeds compared to ASUS and HP, which may impact its efficiency in scenarios requiring random access. **Random Write** Performance: Acer Aspire A515-56 performs well at higher block sizes but falls short at smaller ones.
HP Laptop 15s-fq5xxx	**CPU Technology:** 10nm **Number of Cores**: 2P+8E **Core Speed:** 4090.00 MHz **Bus Speed:** 99.76 MHz **RAM Size:**	**Single - Core:** 2029 **Multi - Core:** 6005	**High CPU Load:** CPU: 13% Memory: 32% Disk: 3% Network: 0% **Mixed Load:** CPU: 12% Memory: 32% Disk: 4% Network: 0%	**Sequential Read** Performance: HP Laptop 15s-fq5xxx performs well across all block sizes, maintaining high read speeds, particularly at 32 KB (1892.66 MB/s) and 16 KB (1913.96 MB/s). **Sequential Write** Performance: HP Laptop 15s-fq5xxx leads with the highest sequential write speeds at 32 KB (2527.83

	32 GBytes		**Low CPU Load:** CPU: 2% Memory: 30% Disk: 5% Network: 0%	MB/s) and performs consistently well across different block sizes. **Random Read** Performance: HP Laptop 15s-fq5xxx also delivers strong random read performance, especially at higher block sizes (32 KB: 1819.16 MB/s). **Random Write** Performance: HP Laptop 15s-fq5xxx leads again in random write performance at 32 KB (2423.92 MB/s) and maintains good speeds across other block sizes.
ASUS Zenbook S13 OLED	**CPU Technology :** 6nm **Number of Cores**: 8 **Core Speed :** 1393.61 MHz **Bus Speed :** 99.49 MHz **RAM Size :** 16 GBytes	**Single - Core :** 1102 **Multi - Core :** 6125	**High CPU Load:** CPU: 5% Memory: 47% Disk: 2% Network: 1% **Mixed Load:** CPU: 7% Memory: 46% Disk: 3% Network: 1% **Low CPU Load:** CPU: 2% Memory: 42% Disk: 1% Network: 1%	**Sequential Read** Performance: ASUS Zenbook S13 OLED exhibits the highest sequential read speeds at 32 KB (2548.21 MB/s) and 4 KB (611.317 MB/s), though it drops for intermediate block sizes. **Sequential Write** Performance: ASUS Zenbook S13 OLED has the lowest sequential write performance, particularly evident at 4 KB (250.595 MB/s), making it less ideal for write-heavy operations. **Random Read** Performance: ASUS Zenbook S13 OLED shines with the highest random read speeds, notably at 32 KB (2583.67 MB/s) and 4 KB (610.731 MB/s). **Random Write** Performance: ASUS Zenbook S13 OLED shows lower random write speeds, especially noticeable at 4 KB (248.885 MB/s).

2.4 Discussion of Result

CPU-Z (System Info)

Acer Aspire 5, with its Intel Core i5 processor, is built on a 10 nm process technology and feature 4 cores. This setup is adequate for basic to moderate computing tasks, such as web browsing, office applications and light multitasking. The 8 GB of RAM supports these activities, but the lower core count and core speed mean it may struggle with more intensive applications or multitasking scenarios. This suitable for general productivity and everyday use but not ideal for demanding workloads. **HP Laptop 15s-fq5xxx**, features a more advanced Intel Core i7 processor with a unique combination of 2 performance cores and 8 efficiency cores. The high core speed (4090.00 MHz) and large RAM (32 GB) make it well-suited for demanding tasks, such as software development, data analysis and multimedia editing. The combination of performance and efficiency cores also provides a balance between power consumption and performance. This is high efficiency for multitasking and resource-intensive applications, making it a versatile choice for professionals. **ASUS Zenbook S13 OLED**, powered by an AMD Ryzen 7 processor, uses a 6nm process technology with 8 cores. This configuration provides excellent performance for both single-threaded and multi-threaded applications. The 16 GB of RAM is sufficient for most professional tasks and heavy multitasking. The advanced 6nm technology suggests better energy efficiency and performance density compared to the 10nm process in the other laptops. This offers the best overall performance and efficiency among the three, making it the **ideal choice for power users and professionals**.

Geekbench (CPU Stress Test)

Acer Aspire 5 shows lower single-core and multi-core scores among the three laptops. This indicates that it is sufficient for basic computing tasks but may not perform well in more demanding scenarios that require significant processing power or multitasking capabilities. The lower cores are reflective of its fewer cores and lower overall performance. **HP Laptop 15s-fq5xxx** achieve the highest single-core score and a solid multi-core score. This suggest strong performance in tasks that require fast, single-threaded processing, as well as good capability for handling multitasking and parallel processing efficiently. This makes it a strong contender for users who required both high single-thread and multi-thread performance. **ASUS Zenbook S13 OLED** has the highest multi-core score, indicating its superior performance in multi-threaded tasks. While its single-core score is slightly lower than the Acer Aspire 5, its multi-core performance makes it ideal for applications that can leverage multiple cores, such as video rendering and complex simulations.

Task Manager (Active Program and Resource Adjustment)

ASUS ZenBook S13 OLED offers the best overall performance, thanks to its advanced 6nm CPU technology which appear to be more efficient than a 10nm process for CPU technology because smaller process sizes pack more transistors into the chip, resulting in better performance and lower power consumption. Other than that ,8 cores and 16GB of RAM, providing excellent CPU efficiency and moderate memory usage across all workloads. **HP Laptop 15s-fq5xxx** also performs well, especially with its substantial 32GB RAM, making it highly efficient in memory management but it has higher CPU usage compared to ASUS under high load. **Acer Aspire 5**, while capable, shows higher CPU and memory usage due to its 10nm 4-core CPU and 8GB RAM, making it less efficient under heavy loads compared to the other two laptops. Thus, **ASUS Zenbook S13 OLED** as leaders in performance and efficiency among these models.

MiniTools (Disk Benchmark)

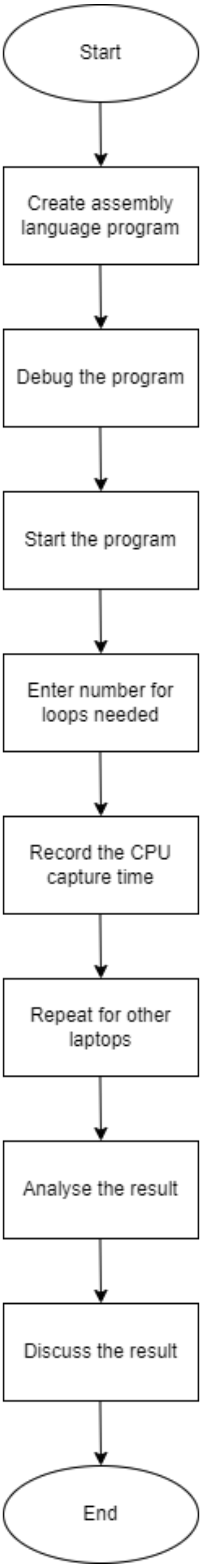
HP Laptop 15s-fq5xxx has the best overall disk performance as it offers superior write performance and maintains strong read speeds, making it versatile for both read and write-intensive applications.

ASUS Zenbook S13 OLED excels in read operations, both sequential and random, making it suitable for tasks demanding high read speeds but lags in write performance, especially at smaller block sizes.

Acer Aspire 5, while not leading in any category, offers a balanced performance suitable for general usage but may not be ideal for high-performance tasks.

3.0 Part 2

3.1 Flowchart of Project Part 2



3.2 Coding and Implementation

```
TITLE COA Project

; Author:
; 1. Evelyn Goh Yuan Qi
; 2. Joanne Ching Yin Xuan
; 3. Lim Yu Han

; Date: 03 July 2024

include Irvine32.inc

.data
msgWelcome BYTE "Welcome to CPU Benchmark Program", 0dh, 0ah, 0
msgAlgorithm BYTE "Benchmark CPU time Using Equation y = (11*x^3) + (17*x^2) + (13*x) + 20", 0ah,
0dh
BYTE "(with delay coef1, coef2, coef3, coef4 = 11, 17, 13, 20 msec)", 0dh, 0ah, 0
msgEnterN BYTE "Enter Number of Looping (N) = ", 0
msgProgress BYTE "CPU time Stress Test in progress...", 0dh, 0ah, 0
msgResult BYTE "Result: ", 0dh, 0ah, 0
msgFirstCapture BYTE "First Capture Execution time in millisecond: ", 0
msgSecondCapture BYTE "Second Capture Execution time in millisecond: ", 0
msgDiffTime BYTE "Different Execution time in millisecond: ", 0
msgSum BYTE "Value of Sum from the Stress Test (polynomial) = ", 0
msgPrompt BYTE "Press 'y' to continue or 'n' to exit the benchmark: ", 0
msgBye BYTE "Thank you ... BYE!!", 0dh, 0ah, 0

coef1 DWORD 11; a = 11
coef2 DWORD 17; b = 17
coef3 DWORD 13; c = 13
coef4 DWORD 20; d = 20

max_loop DWORD ?
sum WORD ?
capture_msec_before DWORD ?
capture_msec_after DWORD ?
elapsed_time DWORD ?

.code
main PROC
; Display welcome message
call clrscr
mov edx, offset msgWelcome
call WriteString
call crlf

; Display the algorithm
mov edx, offset msgAlgorithm
call WriteString
call crlf

; Prompt for max loop value
mov edx, offset msgEnterN
call WriteString
call ReadDec
mov max_loop, eax

; Display progress message
mov edx, offset msgProgress
call WriteString
call crlf

; Capture time before starting the loop
call GetMseconds
mov capture_msec_before, eax
```

```
; Initialize sum to 0
mov sum, 0

; Set loop counter
mov ecx, max_loop
mov ebx, 1

calc_loop:
; Calculate y = (coef1 * x ^ 3) + (coef2 * x ^ 2) + (coef3 * x) + coef4
; coef1* x ^ 3
mov eax, coef1
call Delay
mul bx
mul bx
mul bx
mov dx, ax; Store in dx for adding later

; coef2* x ^ 2
mov eax, coef2
call Delay
mul bx
mul bx
add dx, ax; Add previous result

; coef3* x
mov eax, coef3
call Delay
mul bx
add dx, ax; Add previous result

; coef4
mov eax, coef4
call Delay
add dx, ax; Add coef4

; Add y to sum
add sum, dx

; Increment loop counter
inc ebx
loop calc_loop

; Capture time after completing the loop
call GetMseconds
mov capture_msec_after, eax

; Calculate elapsed time
mov eax, capture_msec_after
sub eax, capture_msec_before
mov elapsed_time, eax

; Display results
mov edx, offset msgResult
call WriteString
call crlf

mov edx, offset msgFirstCapture
call WriteString
mov eax, capture_msec_before
call WriteDec
call crlf

mov edx, offset msgSecondCapture
call WriteString
mov eax, capture_msec_after
call WriteDec
call crlf

mov edx, offset msgDiffTime
```

```
call WriteString
mov eax, elapsed_time
call WriteDec
call crlf

mov edx, offset msgSum
call WriteString
mov ax, sum
call WriteDec
call crlf

; Prompt to continue or exit
mov edx, offset msgPrompt
call WriteString
call ReadChar

cmp al, 'y'
je main

mov edx, offset msgBye
call WriteString

; Exit program
exit
main ENDP
END main
```

3.3 Implementation Result

Computer Type	**No of Loop (N)**	**Capture_msec (before)**	**Capture_msec (after)**	**CPU_msec (different)**
Acer Aspire 5	50	999977	1004667	4690
	100	1028282	1037591	9309
	200	1063048	1081688	18640
	500	1113000	1159582	46582
	700	1181983	1247264	65281
	1000	1294862	1387984	93122
HP Laptop 15s-fq5xxx	50	43171501	43176224	4723
	100	43294987	43304329	9342
	200	43368293	43387096	18803
	500	43434571	43481681	47110
	700	43526607	43592072	65465
	1000	43641527	43734931	93404
ASUS Zenbook S13 OLED	50	64761981	64766709	4728
	100	64838823	64848233	9410
	200	64897692	64916448	18756
	500	64963217	65010237	47020
	700	65342729	65408408	65679
	1000	65479047	65572837	93790

3.4 Analysis of Result

- Acer Aspire 5

  The CPU time (**CPU_msec**) increases as the number of loops (**No of Loop**) increases. The difference in capture time before and after (**Capture_msec (before)**) and (**Capture_msec (after)**) also increases with the number of loops. The trend shows that the processing time roughly scales with the number of loops, indicating that the CPU handles more iterations by taking proportionally more time.

- HP Laptop 15s-fq5xxx

  Similar to Acer Aspire 5, the CPU time increases with the number of loops. The capture times before and after also increase proportionally to the number of loops. The differences in capture time before and after are relative similar to those observed in the Acer Aspire 5, suggesting a consistent increase in processing time with increased workload.

- Asus Zenbook S13 OLED

  The trend of increasing CPU time with more loops continues. The capture time before and after execution show a consistent increase with the number of loops. The difference in capture time follow a pattern similar to the other two laptops, indicating a proportional relationship between the number of loops and the processing time.

3.5 Discussion of Result

The implementation results for the CPU testing program across three different laptops, including the Acer Aspire 5, HP Laptop 15s-fq5xxx, and ASUS Zenbook S13 OLED, provide a complete perspective of their performance under various workloads. The results reveal a clear pattern in which CPU time, represented by the difference in milliseconds, increases as the number of loops grows. This pattern is consistent across all three devices, demonstrating that each laptop's processing time varies proportionally with the workload.

For the Acer Aspire 5, the CPU time difference begins at 4690 milliseconds for 50 cycles and increases to 93122 milliseconds for 1000 loops. This huge rise indicates that, while the laptop can handle greater workloads, the overhead becomes increasingly noticeable as the number of repeats increases. Similarly, the CPU time differential for the HP Laptop 15s-fq5xxx increases from 4723 milliseconds for 50 cycles to 93404 milliseconds for 1,000 loops. This laptop's performance is equivalent to the Acer Aspire 5, with somewhat lower CPU time disparities at larger loop counts. The Asus Zenbook S13 OLED continues this trend, although with slightly lower CPU time disparities throughout all rounds, beginning at 4728 milliseconds for 50 cycles and increasing to 93790 milliseconds for 1000 loops. This suggests that this laptop may be more efficient at managing growing workloads, possibly due to improved hardware specs or increased system performance.

In a nutshell all three laptops show predictable scaling in processing time with higher workloads. However, the Asus Zenbook S13 OLED looks to handle the workload more efficiently, making it an excellent choice for computationally intensive activities.

4.0 Conclusion